An Internship Report

On

Vortex Ring Generator

2D Supersonic Wind Tunnel Design

Under the supervision of

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Executive Summary

This report details a research internship conducted at the Indian Institute of Science (IISc), focusing on experimental aerodynamics and propulsion systems. The internship was focused on three primary works: supersonic facility design, bio-inspired propulsion diagnostics, and compressor stability analysis. Initial efforts focused on the design of a 2D supersonic wind tunnel, utilizing the Method of Characteristics (MoC) to define nozzle contours for uniform test-section of Mach 2.5; 80*100mm test section. In the domain of experimental fluid mechanics, technical support was provided for a Particle Image Velocimetry (PIV) study investigating jellyfish-inspired propulsion. This involved the setup of vortex generators to simulate propulsive wake structures and the subsequent processing of flow-field data using PIVlab to quantify vortex shedding and thrust characteristics. Complementing these experimental tasks, an analytical study was performed on axial compressor surge within jet engines, exploring the fluid dynamic instabilities that lead to flow reversal and performance degradation. This internship provided comprehensive exposure to both high speed aerodynamic design and the application of advanced laser based diagnostics in complex, bio-inspired flows.

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Finally, we extend our sincere appreciation to the **Department of Mechanical and Aerospace Engineering, Pulchowk Campus**, for providing us the opportunity to undertake this internship as part of our final-year academic experience. The institutional support and resources provided made this opportunity possible.

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Sincerely,
Amogh Adhikari
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Chapter 1: Internship Overview

1.1 Host Organization

The Indian Institute of Science (IISc), established in 1909, stands as India's prestigious institution for advanced scientific and technological research and education. Located in Bengaluru, it is globally recognized for its academic environment and its pioneering contributions to aerospace engineering and fluid mechanics. The Department of Mechanical Engineering, one of the oldest and most distinguished at the Institute, was established in 1945. Today, the department is led by its Chairperson and my supervisor, Dr. G. S. Raghuraman, who maintains a focus on technical inquiry and steady innovation.

This emphasis on quality is reflected in the QS World University Rankings, where IISc consistently places within the top 100-150 globally for Mechanical and Aerospace Engineering. For an intern, the department provides a practical environment to work with experienced faculty and contribute to research in propulsion and engineering science.

1.2 Internship Objectives and Scope

- Design a 2D supersonic wind tunnel nozzle using the Method of Characteristics (MoC) to achieve uniform exit flow at Mach 2.5.
- Assist in Particle Image Velocimetry (PIV) setup for a study on jellyfish-inspired propulsion using vortex generators.
- Process experimental flow-field data using PIVlab to quantify vortex shedding and thrust characteristics.
- Study fundamental gas dynamics to bridge theoretical calculations with practical facility design and turbomachinery analysis.

Chapter 2: Background and Theory

2.1 Centrifugal Compressor Fundamentals

Centrifugal compressors are essential components in systems such as turbochargers and small gas turbines, owing to their ability to produce a high pressure ratio in a single stage. While they are powerful and efficient, their performance is constrained by operational limits defined by aerodynamic instabilities, namely stall, surge, and choking.

Stall Stall is a condition in which the airflow through the compressor becomes locally unstable due to boundary layer separation on the blade surfaces. This instability leads to increased pressure fluctuations within the system and a reduction in compressor efficiency.

Surge Surge is a severe form of instability characterized by a complete breakdown of steady flow, resulting in periodic reversal of airflow through the compressor. It is accompanied by a sudden drop in compressor performance and can cause significant mechanical damage due to large pressure and flow oscillations.

Stall and Surge Limit Both stall and surge have damaging effects on the compressor and must be avoided during operation. These phenomena define the low-flow boundary of the compressor's stable operating range.

Choking Choking occurs when the flow through the compressor reaches its maximum capacity, typically when the flow velocity approaches sonic conditions at the narrowest section of the flow path. In this state, the compressor cannot increase the mass flow rate further, even if the downstream (back) pressure is reduced.

2.2 Vortex Ring Fundamentals

A vortex ring is a fundamental fluid mechanical phenomenon commonly identified as the main flow feature in starting jets, volcanic eruptions, and the propulsive actions of aquatic creatures such as squids and jellyfish, as well as the discharge of blood from the left atrium to the left ventricular cavity in the human heart [1]. In a laboratory setting, it is typically generated by a piston pushing a column of fluid of length L through a nozzle of diameter D, which results in the separation of a

boundary layer at the edge and its subsequent spiral roll-up [1]. Physically, a fully formed vortex ring consists of a "vortex bubble" that contains both the fluid originally ejected from the nozzle and ambient fluid that has been entrained during the formation process [2].

Formation Number

The **formation number** is a non-dimensional time scale, also referred to as the stroke ratio L/D, that characterises the vortex ring formation process. It represents the critical point at which a vortex ring stops entraining energy and circulation from the generating jet and "pinches off", resulting in a leading vortex ring followed by a trailing jet [1].
Gharib et al. [1] demonstrated experimentally using Digital Particle Image Velocimetry (DPIV) that this formation number universally lies in the range of 3.6–4.5 for a broad range of flow conditions, including variations in nozzle geometry, exit boundary conditions, and piston velocity programs. Specifically, for a given geometry, the circulation of a vortex ring Γ depends on the history of the piston velocity $u_p(t)$, nozzle diameter D, kinematic viscosity ν, and discharge time t. The non-dimensional formation time is defined as:

$$\frac{L}{D} = \frac{\bar{U}_p\, t}{D} \tag{2.1}$$

where $\bar{U}_p$ is the running mean of the piston velocity defined as:

$$\bar{U}_p = \frac{1}{t}\int_0^t u_p\, dt \tag{2.2}$$

The model proposed by Gharib et al. [1] is based on the hypothesis that the limiting stroke $(L/D)_{\text{lim}}$ occurs when the apparatus is no longer able to deliver energy at a rate compatible with the requirement that a steadily translating vortex ring has maximum energy with respect to impulse-preserving iso-vortical perturbations. This leads to the non-dimensional energy parameter:

$$\alpha \equiv \frac{E}{I^{1/2}\,\Gamma^{3/2}} \tag{2.3}$$

where E is the kinetic energy and I is the impulse of the vortex ring. The formation number is reached when $\alpha_{\text{piston}}(t) < \alpha_{\text{lim}}$, where $\alpha_{\text{lim}} \approx 0.33$ was measured experimentally for the axis-touching vortex rings studied.

Circulation

The **circulation** Γ of a vortex ring is a measure of the rotational strength of the vortex and is defined as the line integral of the velocity field around a closed contour enclosing the vortex core, or equivalently as the surface integral of vorticity ω over the area A enclosed:

$$\Gamma = \oint \mathbf{u} \cdot d\mathbf{l} = \iint_A \omega_z \, dA \tag{2.4}$$

Using the boundary layer assumption, the vorticity flux from the nozzle is approximately $U_p^2/2$, giving the circulation accumulated up to time t as [1]:

$$\Gamma(t) = \frac{1}{2} \overline{U_p^2} \, t \tag{2.5}$$

Gharib et al. [1] showed that the maximum circulation a vortex ring can attain is equivalent to the total circulation discharged from the nozzle up to the formation number. Beyond this point, any additional vorticity from the trailing jet is not entrained into the leading vortex ring. Furthermore, Das et al. [3] showed that the introduction of flexible flaps at the channel exit redistributes the vorticity laterally without significantly changing the total final circulation compared to the rigid exit case.

Impulse and Thrust

The **fluid impulse** I of the flow generated by a vortex ring generator is a key quantity linking vortex ring formation to propulsive performance. The impulse is defined as the time integral of the force F applied to generate the flow [3]:

$$\mathbf{I} = \int_0^\tau \mathbf{F} \, dt \tag{2.6}$$

For a two-dimensional flow field, this can be evaluated directly from the measured vorticity field as [3]:

$$I_x = \rho \int_A y \, \omega_z \, dA \tag{2.7}$$

where ρ is the fluid density, y is the lateral coordinate, and ω_z is the out-of-plane vorticity component evaluated over the area A encompassing all vorticity.

Through a control volume analysis, Krueger and Gharib [2] showed that the total impulse of a starting jet has two components:

$$I(t) = I_U(t) + I_p(t) \tag{2.8}$$

where $I_U(t)$ is the momentum flux contribution:

$$I_U(t) = \rho \int_0^t \int_A u_j^2(r, t) \, dS \, dt \tag{2.9}$$

and $I_p(t)$ is the **pressure impulse**, defined as the contribution from nozzle exit over-pressure:

$$I_p(t) = \int_0^t \int_A [p(r,t) - p_\infty]\, dS\, dt \tag{2.10}$$

Krueger and Gharib [2] demonstrated experimentally that this pressure impulse contribution can be as large as 42% of the total impulse for isolated vortex rings, a contribution entirely neglected by the standard slug model. Furthermore, the **average thrust** during a pulse is defined as:

$$\bar{F}_p = \frac{I}{t_p} \tag{2.11}$$

where t_p is the pulse duration. Krueger and Gharib [2] showed that $\bar{F}_p/(\rho A U_{\max}^2)$ is maximised by pulses near the conditions that generate vortex rings of maximum circulation without leaving a trailing jet, directly linking the formation number to optimal propulsive performance.

The final impulse can also be expressed as the product of the total circulation Γ and an effective lateral vorticity spacing y_{eff} [3]:

$$y_{\text{eff}} = \frac{\int y\, \omega_z\, dA}{\Gamma} \tag{2.12}$$

This effective lateral separation distance represents the distance between the centroid of vorticity on each side of the axis of symmetry. Das et al. [3] showed that flexible flaps at the nozzle exit increase y_{eff} by spreading the vorticity centroids laterally, thereby increasing the final fluid impulse to approximately twice the rigid exit value across a wide range of non-dimensional flexural rigidity EI^* values.

Actuator Types

Two primary actuator types are commonly used to generate vortex rings in laboratory settings:

1. **Piston -Cylinder Mechanism:**
   The most widely used actuator for controlled vortex ring generation [1–3]. A piston is driven through a cylindrical or rectangular duct by a servo motor combined with a lead screw mechanism. The servo motor is controlled by a Programmable Logic Controller (PLC), enabling precise programming of the piston velocity profile $U_p(t)$. The stroke length L and duct diameter or width D (or d for rectangular channels) are the primary geometric parameters. This mechanism allows systematic variation of the stroke ratio L/D and hence direct control over whether a single vortex ring or a vortex ring with a trailing jet is produced.

2. **Diaphragm Actuator:**
   An alternative actuator in which a flexible diaphragm or membrane is deflected to expel a

puff of fluid. This configuration is more commonly used in synthetic jet applications and in biomimetic devices inspired by jellyfish propulsion, where the mantle deformation resembles a diaphragm action. The key advantage over the piston mechanism is the inherent flexibility of the exit geometry, which can passively modify the jet formation process in a manner analogous to the flexible flap configurations studied by Das et al. [3].

2.3 Supersonic Wind Tunnel Design Using MOC

Supersonic wind tunnels are specialized ground testing facilities designed to generate airflows at speeds exceeding the local speed of sound ($M > 1$). These facilities are essential for simulating the aerodynamic environment encountered by high speed aircraft, missiles, and spacecraft during atmospheric flight.

Working Principle

The fundamental operation of a supersonic wind tunnel relies on the acceleration of gas using a convergent-divergent (De Laval) nozzle [4]. . In a subsonic flow, the gas accelerates as the flow area decreases. To reach supersonic speeds, the duct must converge to a minimum area, known as the throat, where the flow becomes choked at Mach 1 . Beyond the throat, the area diverges, allowing the gas to continue accelerating into the supersonic regime [4].

Structural Components

A standard contoured supersonic nozzle is divided into five primary regions:

- **Contraction Region**: A subsonic section where flow area decreases to accelerate the gas [4].
- **Throat Region**: The minimum cross-sectional area where the flow transitions from subsonic to low supersonic speeds.
- **Initial Expansion Region**: Located immediately downstream of the throat, where centered expansion waves are generated to rapidly increase the Mach number [4] [5].
- **Straightening Section**: A contoured region where the wall slope turns back toward the axis to cancel expansion waves, ensuring parallel flow [4].
- **Test Section**: The final area where the flow is uniform, shock-free, and parallel for experimental measurements [4] [5] [6]

Classification of Facilities

While the sources focus heavily on nozzle design, they distinguish between facility types based on their operation

- **Blowdown Facilities**: These utilize a high-pressure vessel to store air . The tunnel runs intermittently as the total mass of air in the pressure vessel is discharged through the nozzle over a specific time interval [4].
- **Continuous Facilities**: These are designed to maintain a steady pressure ratio over long durations, often required for complex investigations into nozzle performance over a wide flight range [4] [6].

Supersonic Nozzle Design Details

The nozzle geometry is divided into two distinct regions: a subsonic contraction and a supersonic expansion. The design ensures a smooth transition to the throat and a shock-free, uniform flow at the exit.

Convergent Section: Quintic Polynomial To minimize boundary layer separation and ensure a smooth flow transition from the inlet radius R_{inlet} to the throat radius R^*, a fifth-order (quintic) polynomial is employed. This method ensures that the slope and curvature are zero at both the inlet and the throat. The wall coordinates $y(x)$ over the convergence length L_{conv} are defined as:

$$y(x) = R_{inlet} + (R^* - R_{inlet}) \left[10 \left(\frac{x}{L_{conv}} \right)^3 - 15 \left(\frac{x}{L_{conv}} \right)^4 + 6 \left(\frac{x}{L_{conv}} \right)^5 \right] \tag{2.13}$$

Divergent Section: Method of Characteristics (MoC) The supersonic expansion is designed using the Method of Characteristics for a Minimum Length Nozzle (MLN). This technique transforms the non-linear partial differential equations of supersonic flow into ordinary differential equations along characteristic Mach lines (C^+ and C^-).

1. **Prandtl-Meyer Expansion:** The expansion of the flow from $M = 1$ at the throat to the design Mach number M_e is governed by the Prandtl-Meyer function, $\nu(M)$:

$$\nu(M) = \sqrt{\frac{\gamma+1}{\gamma-1}} \tan^{-1} \left(\sqrt{\frac{\gamma-1}{\gamma+1}(M^2-1)} \right) - \tan^{-1}(\sqrt{M^2-1}) \tag{2.14}$$

2. **Compatibility Relations:** Along the characteristics, the following properties remain constant:

- Along C^- (right-running): $K_- = \nu + \theta = \text{constant}$
- Along C^+ (left-running): $K_+ = \nu - \theta = \text{constant}$

Where θ represents the local flow angle.

3. **Minimum Length Geometry:** For a sharp-corner expansion, the maximum wall turning angle θ_{max} is set to half the total expansion angle required for the target exit Mach number:

$$\theta_{max} = \frac{\nu(M_e)}{2} \tag{2.15}$$

Isentropic Verification The final exit radius R_{exit} is verified against the 1D isentropic area-ratio relation to ensure consistency with the design Mach number $M_e = 2.5$:

$$\frac{A}{A^*} = \frac{1}{M_e}\left[\frac{2}{\gamma+1}\left(1 + \frac{\gamma-1}{2}M_e^2\right)\right]^{\frac{\gamma+1}{2(\gamma-1)}} \tag{2.16}$$

2.4 Particle Image Velocimetry (PIV)

Particle Image Velocimetry (PIV), specifically the digital version known as DPIV, is an optical method of flow visualisation used to obtain instantaneous velocity measurements and two-dimensional displacement-vector fields [1, 3]. The technique involves the following components:

1. **Seeding Particles:**
   The flow is seeded with neutrally buoyant tracer particles that follow the flow dynamics without altering them. Silver-coated spherical hollow glass particles with mean diameters ranging from 14 μm to 50 μm are commonly used, as they reflect laser light effectively [1, 2].

2. **Illumination:**
   A thin laser sheet is passed through the central plane of the flow to illuminate the seeding particles.

   - *Laser Type:* Continuous lasers (such as a 1 W green laser) or pulsed lasers (such as an Nd:YAG laser) are used depending on the flow speed and required temporal resolution [2, 3].
   - *Sheet Thickness:* The laser sheet is typically between 2 mm and 3 mm thick, generated using a cylindrical lens [3].

3. **Imaging:**
   A high-speed camera is positioned perpendicular to the laser sheet to record the movement of the illuminated particles.

   - *Equipment:* High-speed cameras such as the Phantom Miro 110 (1 megapixel) or standard CCD (charge-coupled device) cameras are typically used [3].
   - *Frame Rates:* Capture rates typically range from 200 to 350 fps depending on the flow speed [3].
   - *Optimisation:* The frame rate and exposure time are selected to ensure particle displacement of approximately 5–8 pixels between successive images, while avoiding streaking of particle motion.

4. **Data Processing:**
   Captured image sequences are analysed by pairing subsequent images and applying cross-correlation to extract velocity fields.

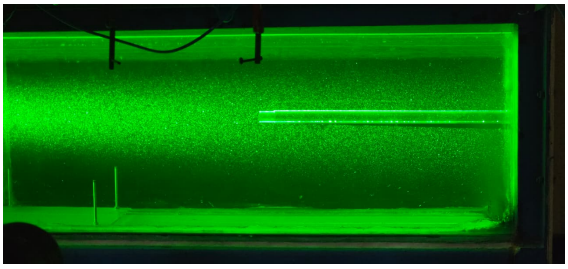
   - *Algorithms:* Adaptive correlation or cross-correlation with a window-shifting algorithm is typically implemented in software such as DynamicStudio (Dantec Dynamics) [3].
   - *Interrogation Windows:* Images are divided into interrogation boxes of 16×16 or 32×32 pixels with 50% overlap to increase spatial vector density [3].
   - *Velocity Accuracy:* Sub-pixel accuracy of approximately 0.1 pixel yields a velocity uncertainty of roughly 1% of the peak measured velocity [3].
   - *Vorticity Accuracy:* Vorticity evaluated using a central-difference scheme carries an uncertainty of less than 10% of the peak vorticity [3].

Chapter 3: Methodology

3.1 PIV System Setup

The PIV system consists of the following components: a water channel/flow channel driven by a piston cylinder actuator, a continuous green laser (532 nm) for illuminating seeding particles and a high speed/CCD camera for image acquistion. The flow is seeded with particles to allow velocity field measurement.

The PLC(Programmable Logic Controller) Controls the water pump and the actuator timing. The camera capture double frame images which are cross correlated to obtain 2D velocity vectors across the measurement plane.



(a) Zoom view of the water channel



(b) Wide view of the setup

Figure 3.1: PIV setup views.

3.2 PIV Experimental Setup

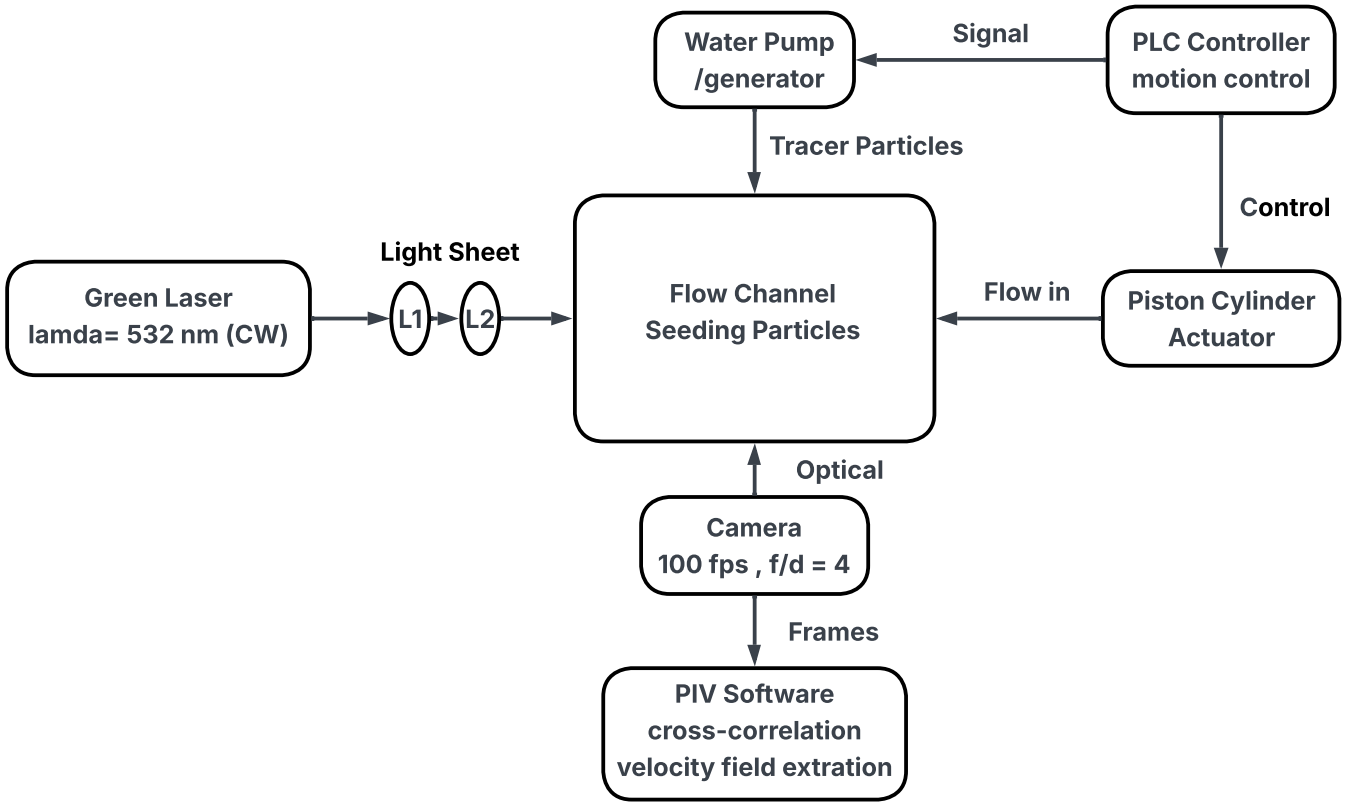


Figure 3.2: Simplified schematic of the PIV experimental setup

3.3 PIV Acquisition Parameters

The PIV system was operated at an acquisition frequency of 100 fps, yielding a temporal resolution of 10 ms per frame. The spatial calibration of the optical setup was determined to be $0.229 mm$/pixel, allowing accurate conversion of image plane displacements into physical velocity vectors. These acquisition parameters were selected to ensure that the particle displacements between consecutive frames remained within the optimal range for cross correlation analysis, while simultaneously resolving the spatial structure of the vortex dipole generated by the oscillating source.

Chapter 4: Results

4.1 2D Supersonic Wind Tunnel Design

Table 4.1: Method of Characteristics (MOC) Numerical Results for $M_e = 2.5$

Point	K^- **(deg)**	K^+ **(deg)**	θ **(deg)**	ν **(deg)**	μ **(deg)**	**Mach**
Throat	0.0000	0.0000	0.0000	0.0000	90.0000	1.0000
Fan-1	6.5206	0.0000	3.2603	3.2603	57.3430	1.1878
Fan-2	13.0412	0.0000	6.5206	6.5206	49.6198	1.3127
Fan-3	19.5618	0.0000	9.7809	9.7809	44.4716	1.4274
Fan-4	26.0824	0.0000	13.0412	13.0412	40.5405	1.5385
Fan-5	32.6030	0.0000	16.3015	16.3015	37.3379	1.6488
Fan-6	39.1236	0.0000	19.5618	19.5618	34.6257	1.7599
CL-1	6.5206	-6.5206	0.0000	6.5206	49.6198	1.3127
CL-2	13.0412	-13.0412	0.0000	13.0412	40.5405	1.5385
CL-3	19.5618	-19.5618	0.0000	19.5618	34.6257	1.7599
CL-4	26.0824	-26.0824	0.0000	26.0824	30.1793	1.9892
CL-5	32.6030	-32.6030	0.0000	32.6030	26.5947	2.2338
CL-6	39.1236	-39.1236	0.0000	39.1236	23.5782	2.5000
Wall-1	45.6442	-6.5206	19.5618	26.0824	30.1793	1.9892
Wall-2	44.3400	-13.0412	15.6494	28.6906	28.6626	2.0849
Wall-3	43.0359	-19.5618	11.7371	31.2989	27.2592	2.1833
Wall-4	41.7318	-26.0824	7.8247	33.9071	25.9525	2.2851
Wall-5	40.4277	-32.6030	3.9124	36.5153	24.7291	2.3905
Wall-6	39.1236	-39.1236	0.0000	39.1236	23.5782	2.5000

The design of the 2D supersonic nozzle was validated through a numerical integration of the governing equations using the Method of Characteristics (MOC). The resulting characteristic mesh, as summarized in Table 4.1, illustrates the evolution of flow properties from the sonic throat to the uniform exit at $M_e = 2.50$.

At the throat, the flow is initialized with a Mach number of 1.00 and a flow angle (θ) of 0°. As the flow enters the divergent section, it undergoes a sharp-corner expansion. This is represented by points **Fan-1 through Fan-6**, where the expansion waves originate. In this region, the Prandtl-Meyer function (ν) increases in equal increments up to 19.56°, causing a corresponding increase in the Mach number to 1.76, while the left-running characteristic constant (K^+) remains zero.

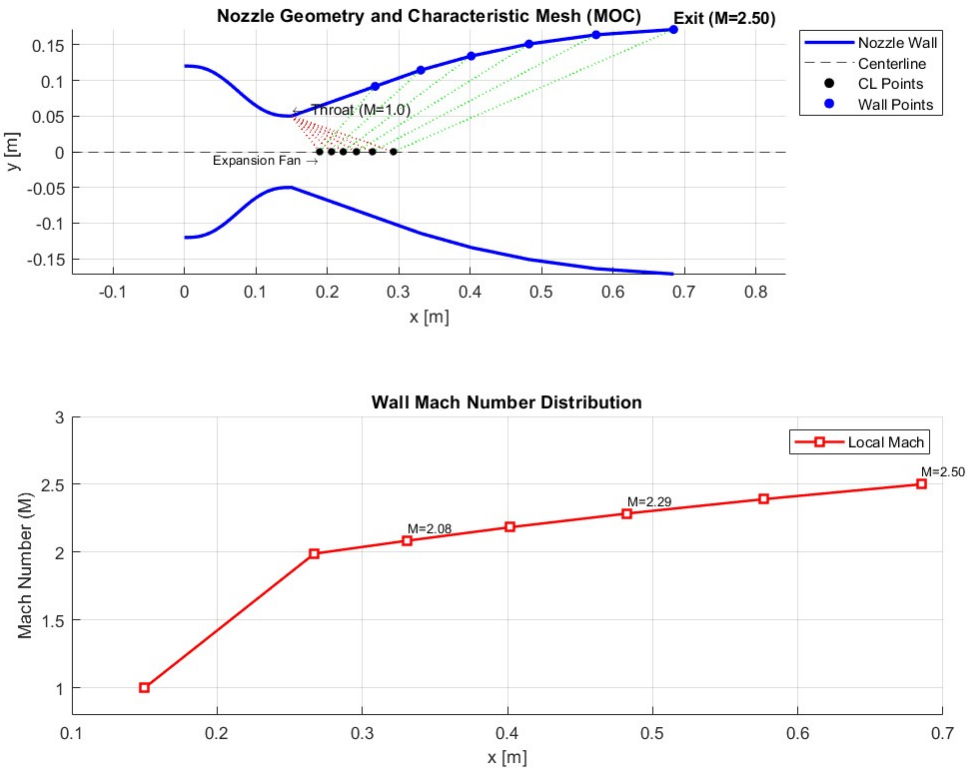


Figure 4.1: Characteristic mesh and wall contour for the $M_e = 2.5$ supersonic nozzle, illustrating the expansion fan and centerline reflections.

The expansion waves then propagate to the nozzle axis, represented by points **CL-1 through CL-6**. To satisfy the physical boundary condition of symmetry at the centerline, the flow angle θ must return to 0°. This reflection process further accelerates the flow; by the final centerline point, the flow reaches the design Mach number of 2.50.

Finally, these reflected waves are used to define the actual wall coordinates at points **Wall-1 through Wall-6**. To achieve a shock-free, parallel exit flow, the wall slope is gradually reduced from its maximum expansion angle of 19.56° back to 0°. The successful convergence of the wall Mach number to 2.50 at a zero-degree flow angle confirms that the nozzle geometry is correctly contoured to provide uniform supersonic flow at the design point.

4.2 PIV Plots and Graphs

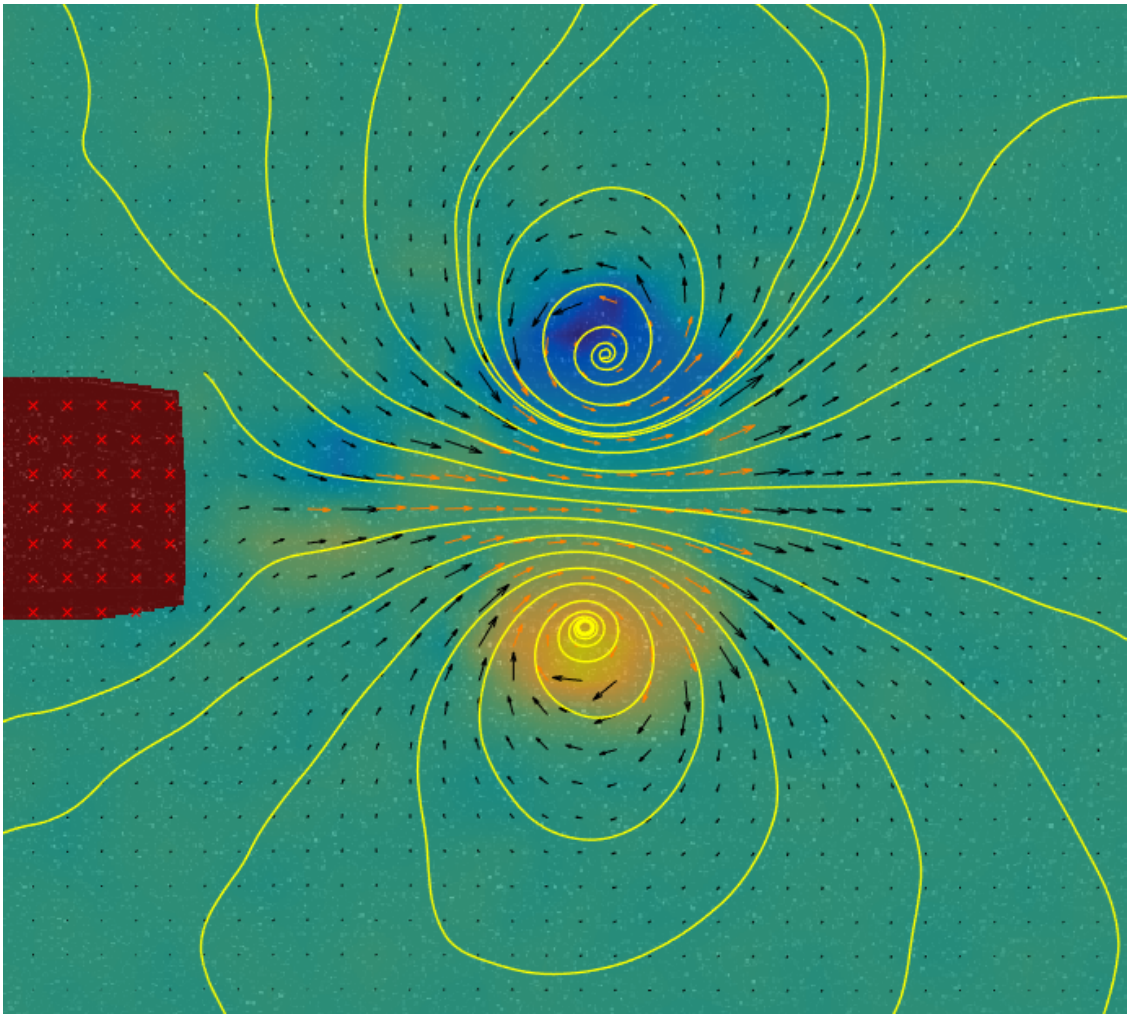


Figure 4.2: Visualization of a Vortex Street Wake using Streamlines and Velocity Vectors.

The yellow streamlines illustrate the instantaneous direction of flow, highlighting the formation of a vortex pair (dipole) in the wake. The background color contours represent vorticity (ω), where the blue region indicates clockwise rotation and the orange/yellow region indicates counter-clockwise rotation. The convergence of streamlines toward the center of these regions identifies the vortex cores, where fluid becomes trapped in a recirculating pattern.

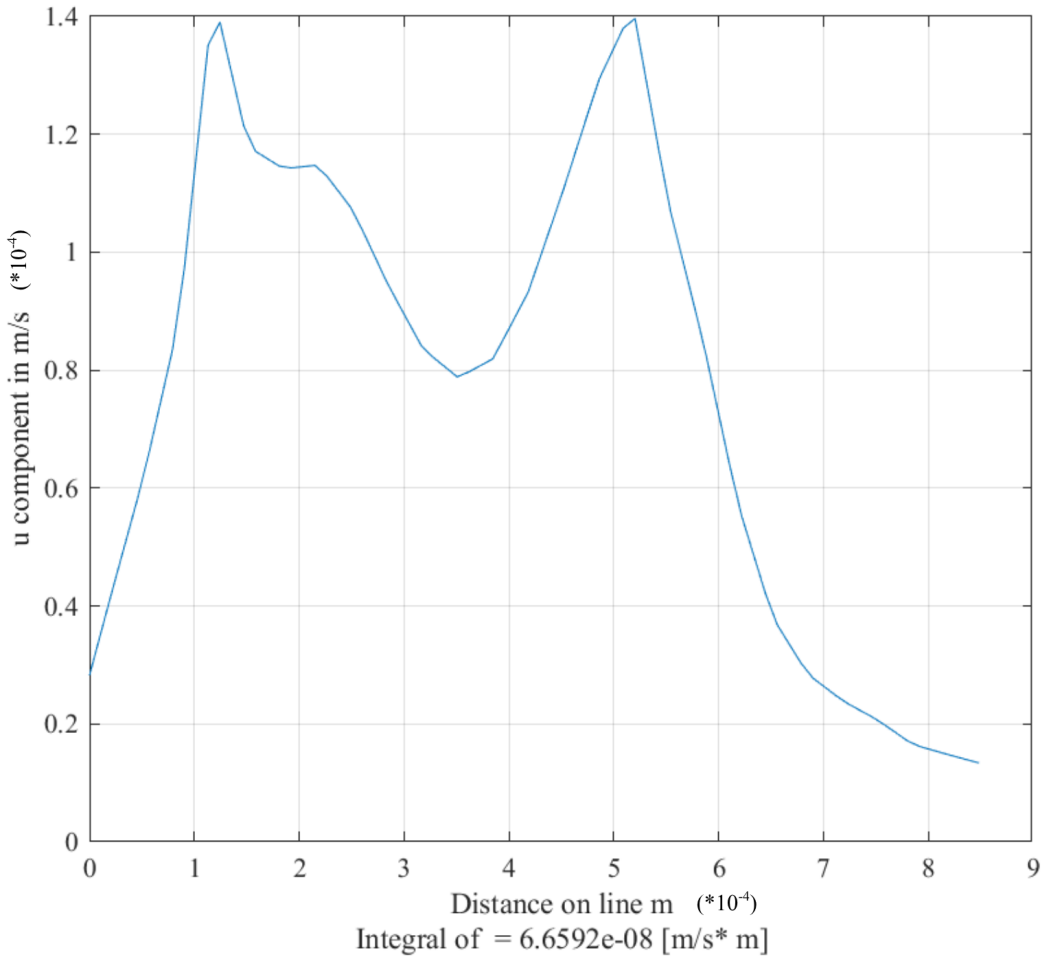


Figure 4.3: U component Velocity Profile

The velocity profile exhibits two distinct peaks along the measurement line. The presence of these two peaks is consistent with the two counter-rotating vortices generated by the oscillating source, where each vortex core produces a local maximum in the horizontal velocity component. The dip between the two peaks indicates the transition zone between the two vortex structures.

The integral of the u-component along the measurement line was calculated as:

$$\int u \, dl = 6.6592 \times 10^{-8} \, [\text{m/s} \cdot \text{m}] \tag{4.1}$$

This value quantifies the net horizontal momentum flux carried by the flow along the sampled line, providing a measure of the overall strength of the horizontal velocity field at this instant in time.

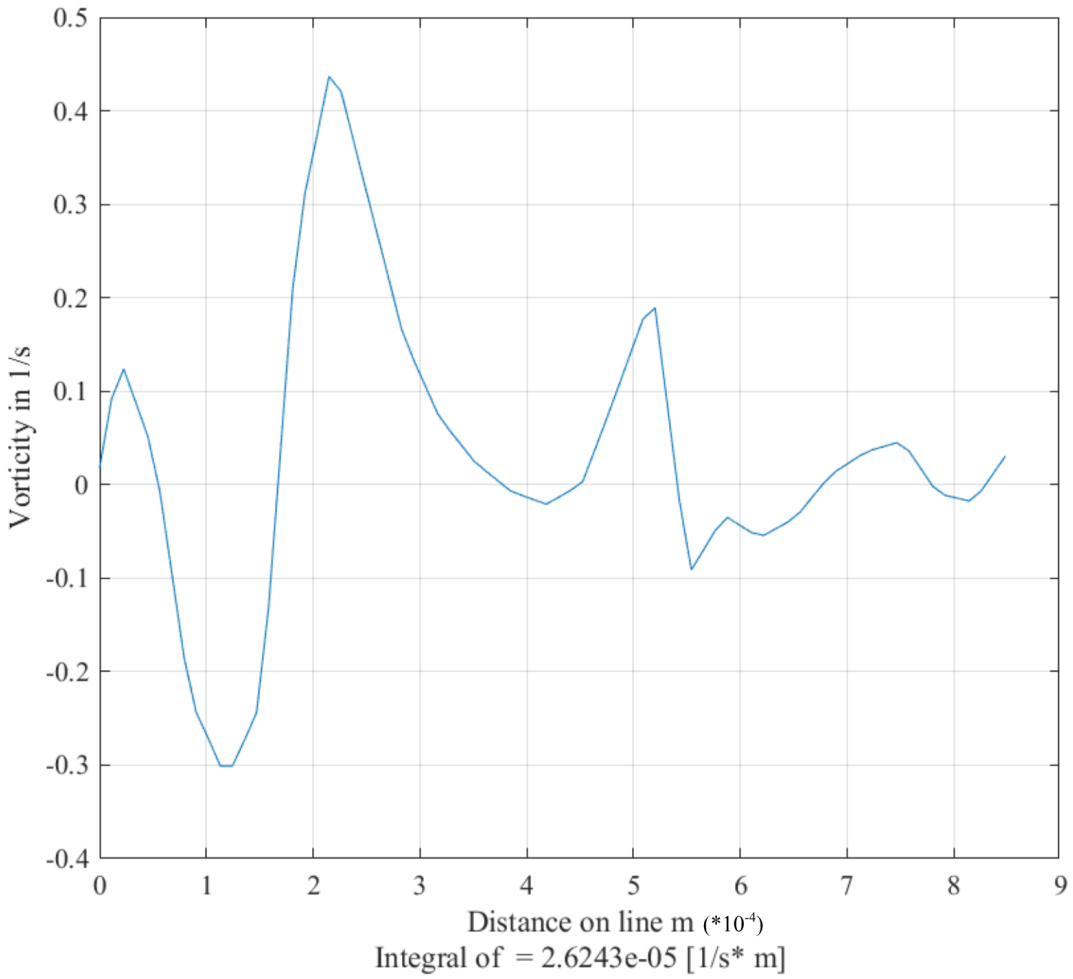


Figure 4.4: Vorticity along horizontal line

The profile begins with a small positive peak of approximately $+0.12\ \mathrm{s}^{-1}$ near the origin, before sharply declining to a negative minimum of about $-0.30\ \mathrm{s}^{-1}$ at approximately 1.2×10^{-4} m, indicating a localized region of clockwise (negative) rotation. This is followed by the most prominent feature of the profile, a sharp positive peak of around $+0.44\ \mathrm{s}^{-1}$ at approximately 2.2×10^{-4} m, representing the strongest region of counter-clockwise (positive) rotation along the line.

The integral of the vorticity along the measurement line is:

$$\int \omega \, dl = 2.6243 \times 10^{-5}\ [\mathrm{s}^{-1} \cdot \mathrm{m}]$$

The net positive integral confirms that counter-clockwise vorticity is dominant along the sampled line.

Chapter 5: Conclusions

A 2D supersonic tunnel was designed using the method of characteristics, and the vortex ring generation experiment was successful. Additionally, post-processing was performed using PIVlab in MATLAB.

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Chapter A: MATLAB Source Code: MOC Nozzle Design

```
% MOC Nozzle Design - Mach 2.5 Exit
% Based on Method of Characteristics (Anderson Ch. 11)
clear; clc; close all;

%% Input Parameters
Me      = 2.5;
N       = 6;
gamma   = 1.4;
R_star  = 0.05;
R_inlet = 0.12;
Lconv   = 0.15;

%% Constants and Setup
nu_e      = pm_deg(Me);
theta_max = nu_e / 2;
d_theta   = theta_max / N;

% Converging Section (Quintic)
xc = linspace(0, Lconv, 300);
t  = xc / Lconv;
yc = R_inlet + (R_star-R_inlet).*(10*t.^3 - 15*t.^4 + 6*t.^5);

%% MOC Table Calculation
n_rows  = 1 + 3*N;
T_theta = zeros(n_rows,1);  T_nu = zeros(n_rows,1);
T_M     = zeros(n_rows,1);  T_mu = zeros(n_rows,1);
T_Km    = zeros(n_rows,1);  T_Kp = zeros(n_rows,1);

% Throat
T_M(1)=1.0; T_mu(1)=90;

% Expansion Fan (Group A)
```

```
Km_A = zeros(1,N);
for i=1:N
    th = i*d_theta;
    row = 1+i;
    T_theta(row)=th; T_nu(row)=th; T_Km(row)=2*th;
    T_M(row) = pm_inv(th); T_mu(row)=mu_deg(T_M(row));
    Km_A(i) = 2*th;
end

% Centerline (Group B)
nu_B = zeros(1,N); M_B = zeros(1,N); mu_B = zeros(1,N);
for i=1:N
    row = 1+N+i;
    T_nu(row) = Km_A(i); T_Kp(row) = -Km_A(i);
    T_M(row) = pm_inv(T_nu(row)); T_mu(row) = mu_deg(T_M(row));
    nu_B(i) = T_nu(row); M_B(i) = T_M(row); mu_B(i) = T_mu(row);
end

% Wall (Group C)
th_C = zeros(1,N); M_C = zeros(1,N);
for i=1:N
    row = 1+2*N+i;
    th_C(i) = theta_max*(N-i)/(N-1);
    T_theta(row) = th_C(i);
    T_nu(row) = th_C(i) + nu_B(i);
    T_M(row) = pm_inv(T_nu(row)); T_mu(row) = mu_deg(T_M(row));
    M_C(i) = T_M(row);
end

%% Coordinates
x_cl = zeros(1,N);
for i=1:N
    phi = 0.5*((i*d_theta - mu_deg(pm_inv(i*d_theta))) + (0 - mu_B(i)))
        ;
    x_cl(i) = Lconv + R_star/abs(tan(deg2rad(phi)));
end

x_w = zeros(1,N+1); y_w = zeros(1,N+1);
```

```
x_w(1)=Lconv; y_w(1)=R_star;
curr_th = theta_max;
for i=1:N
    phi_c = 0.5*(mu_B(i) + (th_C(i)+mu_deg(M_C(i))));
    sC = tan(deg2rad(phi_c));
    sw = tan(deg2rad(curr_th));
    x_w(i+1) = (sC*x_cl(i) - sw*x_w(i) + y_w(i))/(sC - sw);
    y_w(i+1) = sC*(x_w(i+1) - x_cl(i));
    curr_th = th_C(i);
end

%% Result Plotting
figure('Name','Nozzle MOC Analysis','Color','w','Position', [100, 100,
   900, 700]);

% --- Subplot 1: Geometry and Mesh ---
subplot(2,1,1);
hold on; grid on; axis equal;

% Plot Wall and Centerline
plot([xc, x_w], [yc, y_w], 'b', 'LineWidth', 2, 'DisplayName', 'Nozzle
   Wall');
plot([xc, x_w], -[yc, y_w], 'b', 'LineWidth', 2, 'HandleVisibility', '
   off');
yline(0, 'k--', 'LineWidth', 0.5, 'DisplayName', 'Centerline');

% Draw MOC lines (Fan and Reflections)
for i=1:N
    % Right-running waves (Fan to CL) - Hidden from legend
    plot([Lconv, x_cl(i)], [R_star, 0], 'r:', 'LineWidth', 1, '
       HandleVisibility', 'off');

    % Left-running waves (CL to Wall) - Hidden from legend
    plot([x_cl(i), x_w(i+1)], [0, y_w(i+1)], 'g:', 'LineWidth', 1, '
       HandleVisibility', 'off');
end
% Add Points and Labels
```

```
scatter(x_cl, zeros(1,N), 20, 'k', 'filled', 'DisplayName', 'CL Points');
scatter(x_w(2:end), y_w(2:end), 30, 'b', 'filled', 'DisplayName', 'Wall Points');

% Physical Annotations
text(Lconv, R_star*1.2, '\leftarrow Throat (M=1.0)', 'FontSize', 9);
text(x_cl(1), -0.01, 'Expansion Fan \rightarrow', 'FontSize', 8, 'HorizontalAlignment', 'right');
text(x_w(end), y_w(end)*1.1, sprintf('Exit (M=%.2f)', M_C(end)), 'FontWeight', 'bold');

title('Nozzle Geometry and Characteristic Mesh (MOC)');
xlabel('x [m]'); ylabel('y [m]');
legend('Location', 'northeastoutside');

% --- Subplot 2: Mach Distribution ---
subplot(2,1,2);
hold on; grid on;

% Plot Mach curve
plot(x_w, [1, M_C], 'r-s', 'LineWidth', 1.5, 'MarkerFaceColor', 'w', 'DisplayName', 'Local Mach');

% Label specific points on the curve
for i=1:length(M_C)
    if mod(i,2) == 0 || i == length(M_C) % Label every 2nd point to avoid clutter
        text(x_w(i+1), M_C(i)+0.1, sprintf('M=%.2f', M_C(i)), 'FontSize', 8);
    end
end

title('Wall Mach Number Distribution');
xlabel('x [m]'); ylabel('Mach Number (M)');
ylim([0.8, Me + 0.5]);
legend('Location', 'northeast');

```

```
%% Data Export
wall_data = [(0:N)', [1; M_C']];
save('nozzle_results.mat');
fprintf('Design Complete. Exit Mach: %.3f\n', M_C(end));

%% Local Functions
function m = pm_inv(nu_deg)
    nu = deg2rad(nu_deg); m = 2.0;
    g = 1.4;
    for k=1:20
        f = sqrt((g+1)/(g-1))*atan(sqrt((g-1)/(g+1)*(m^2-1))) - atan(
           sqrt(m^2-1)) - nu;
        df = (sqrt(m^2-1)/(1 + (g-1)/2*m^2))/m;
        m = m - f/df;
    end
end

function d = pm_deg(M)
    g = 1.4;
    d = rad2deg(sqrt((g+1)/(g-1))*atan(sqrt((g-1)/(g+1)*(M^2-1))) -
       atan(sqrt(M^2-1)));
end

function mu = mu_deg(M)
    mu = rad2deg(asin(1/M));
end
```

Chapter B: Pressure Rake Design

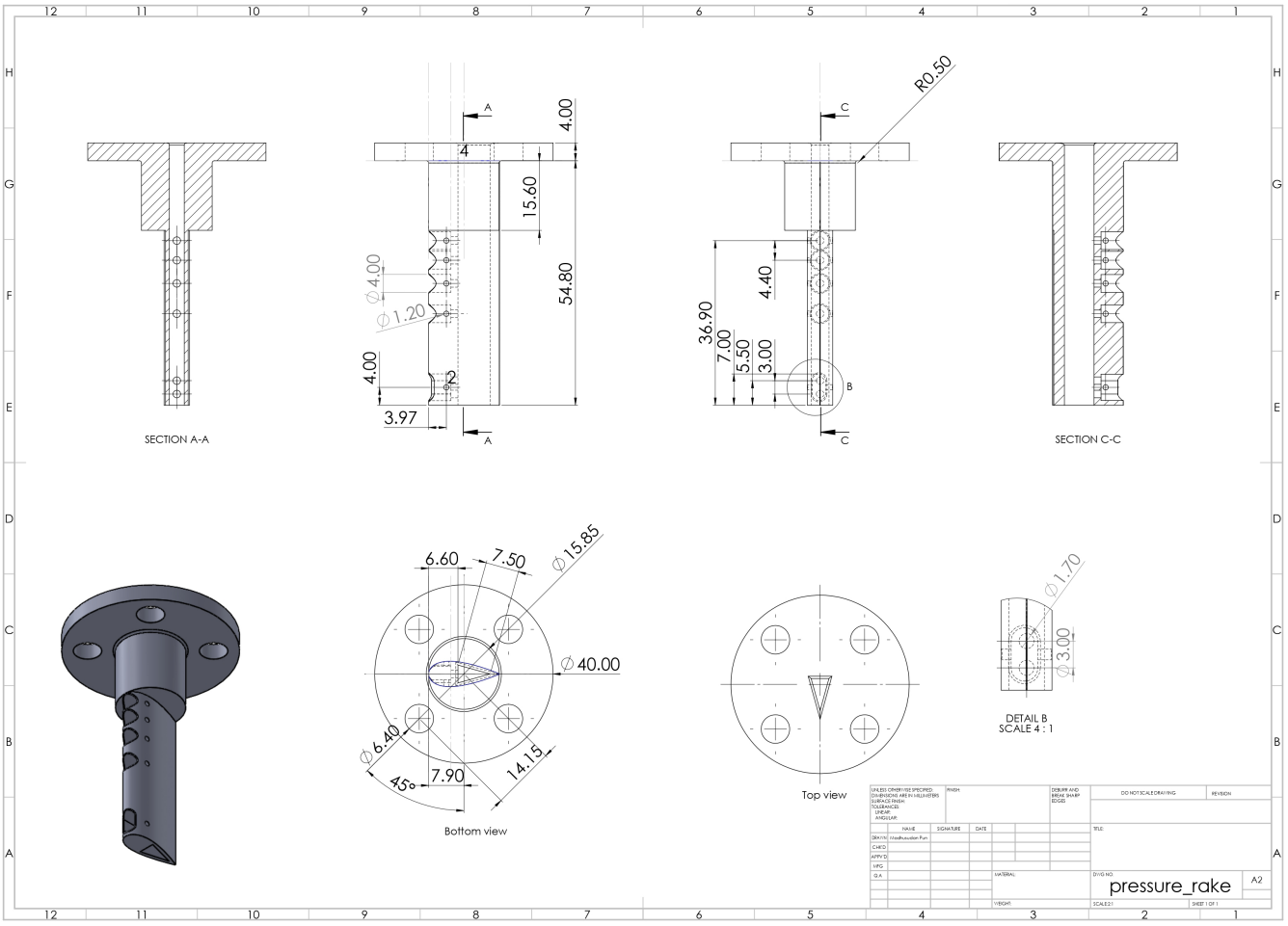


Figure B.1: Pressure Rake